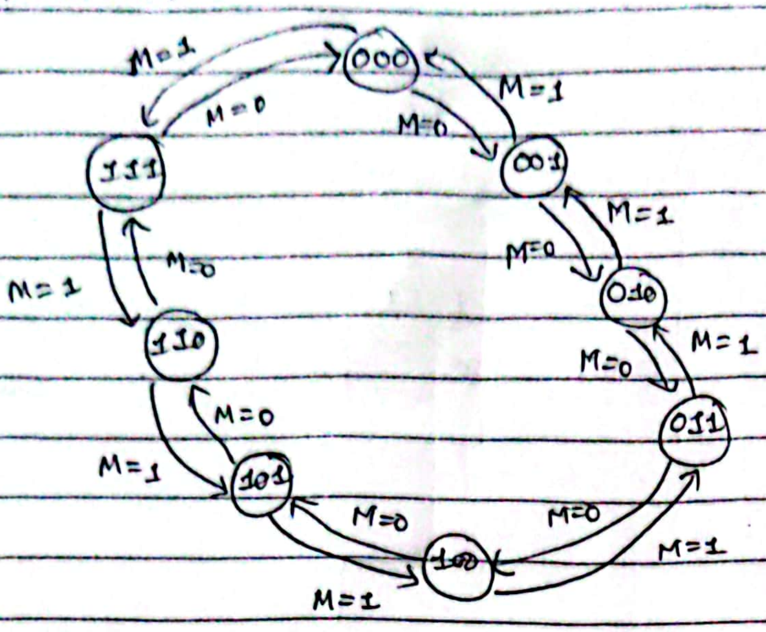
* Design up/down counter (synchronous)

=) Solution:-

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

1) State Table:-



2) State Diagram

Mode	Present state			Next state			Flip Flops		
M	Q_2	Q_1	Q_0	Q_2'	Q_1'	Q_0'	T_2	T_1	T_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

T_2

$MQ_2 \backslash Q_1Q_0$	00	01	11	10
00			1	
01			1	
11	1			
10	1			

$$T_2 = \overline{M} Q_1 Q_0 + M \overline{Q_1}\,\overline{Q_0}$$

T_1

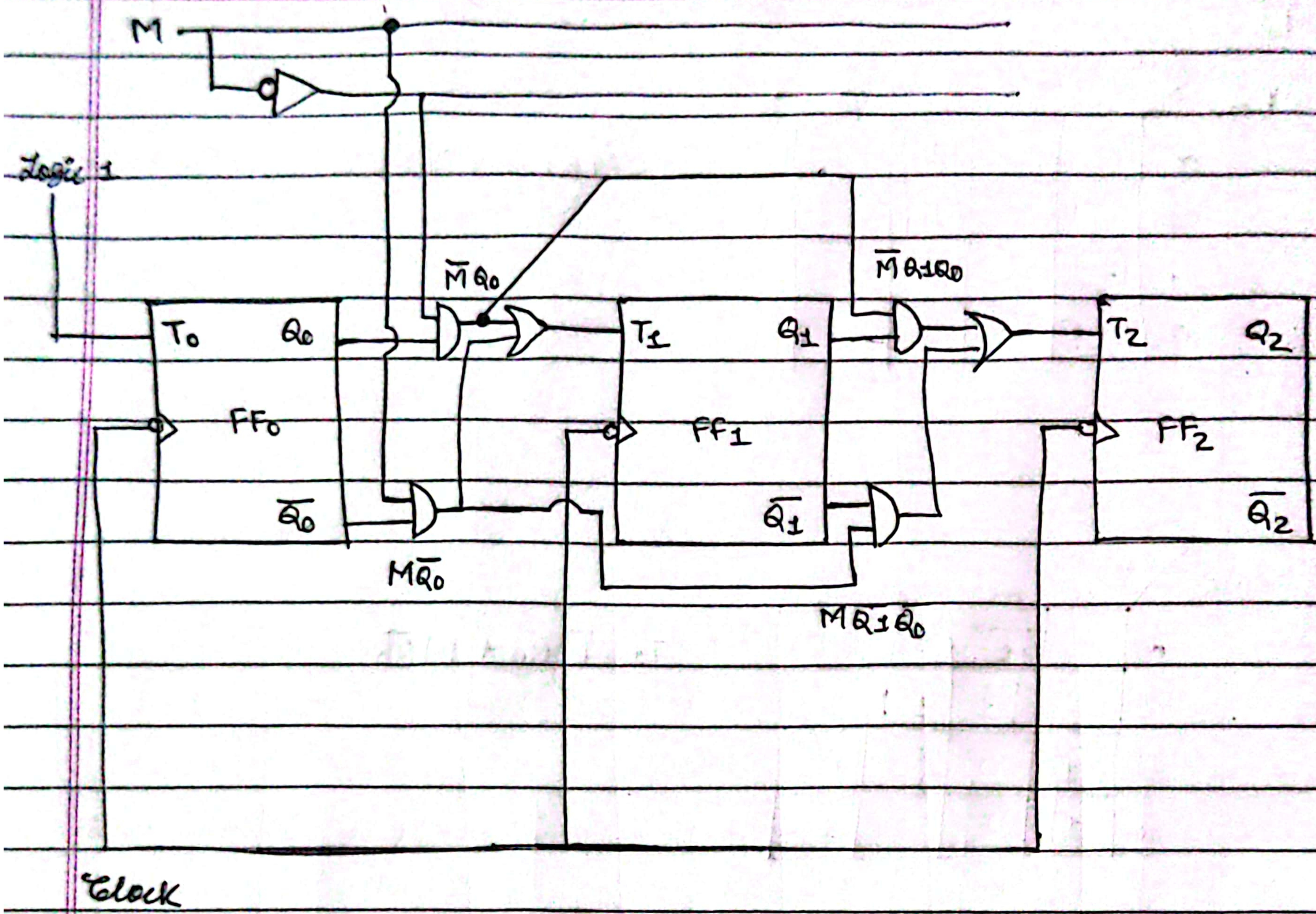
$MQ_2 \backslash Q_1Q_0$	00	01	11	10
00		1	1	
01		1	1	
11	1			1
10	1			1

$$T_1 = \overline{M} Q_0 + M \overline{Q_0}$$

T_0

$MQ_2 \backslash Q_1Q_0$	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$T_0 = 1$$


M
Logic 1
$\overline{M}Q_0$
$\overline{M}Q_1Q_0$
T_0
Q_0
FF_0
$\overline{Q_0}$
$M\overline{Q_0}$
T_1
Q_1
FF_1
$\overline{Q_1}$
MQ_1Q_0
T_2
Q_2
FF_2
$\overline{Q_2}$
Clock